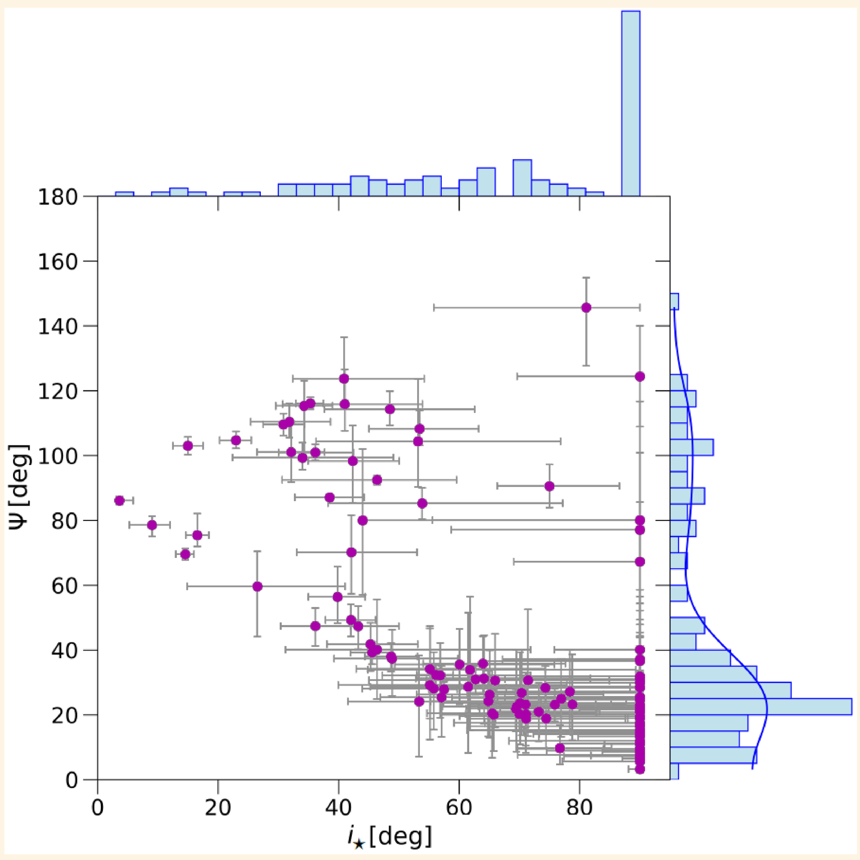
Stellar inclinations and spin-orbit angles from photometry and spectroscopy

1) Why do we need stellar inclinations and spin-orbit angles?

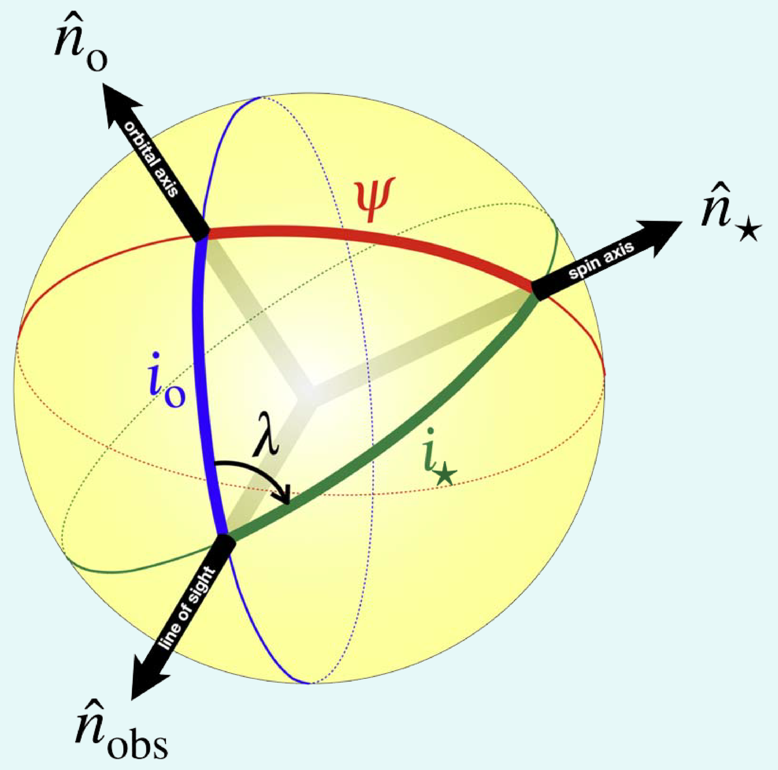
- To date more than 6 000 exoplanets were discovered, but their evolution is still not well constrained [1, 2, 3, 4]
- Characterisation requires information about eccentricity, planetary mass and spin-orbit angle
- Mass and eccentricity are measured during the candidate confirmation process
- But only about 250 sky-projected spin-orbit and 110 true spin-orbit angles are available [2]
- Further characterisation is limited by our understanding of stellar parameters [2] "Know thy star, know thy planet"
- Stellar inclinations are a bottleneck
- Possibility of primordial misalignment of the accretion disk [5]



The observed Ψ-$i_\star$ distribution from [2].

2) Geometry of the problem

- $\hat{n}_o$ - orbital axis
- $\hat{n}_\star$ - stellar rotation axis
- $\hat{n}_{\rm obs}$ - direction to observer
- i_o - orbital inclination, $\approx 90°$ is required for transit observation
- $i_\star$ - stellar inclination
- λ - sky-projected spin-orbit angle
- Ψ - true spin-orbit angle



Relevant geometrical quantities as shown in [1].

- Sky-projected spin-orbit angle λ can be measured during transits from the Rossiter–McLaughlin effect [10,11]
- Need to de-project $\lambda \Rightarrow \Psi$. For this, stellar inclination $i_\star$ is required: $\cos(\Psi) = \sin(i_\star)\cos(\lambda)\sin(i_{\rm orb}) + \cos(i_\star)\cos(i_{\rm orb})$ [1]

3) Deriving stellar inclinations

Stellar inclinations can be derived by combining the stellar rotation period $P_{\rm rot}$, radius $R_\star$ and apparent rotational velocity $v\sin(i_\star)$. [1, 2] Neglecting differential rotation, the relation $\sin(i_\star) \equiv \frac{v_{\rm rot}\sin(i_\star)}{v_{\rm rot}} = \frac{P_{\rm rot}\, v\sin(i_\star)}{2\pi R_\star}$ seems trivial, however:

- Identifying rotation periods can be complicated by additional variability [6]
- Differential rotation changes observed $P_{\rm rot}$ and $v\sin(i_\star)$
- Long and precise photometric series are required
- Source of homogeneous stellar radii is needed (*Gaia* [7], *ARIADNE* [8])
- The values of $v_{\rm rot}$ and $v\sin(i)$ are not independent and a Bayesian approach is required [9]

4) Proposed Photometry Pipeline

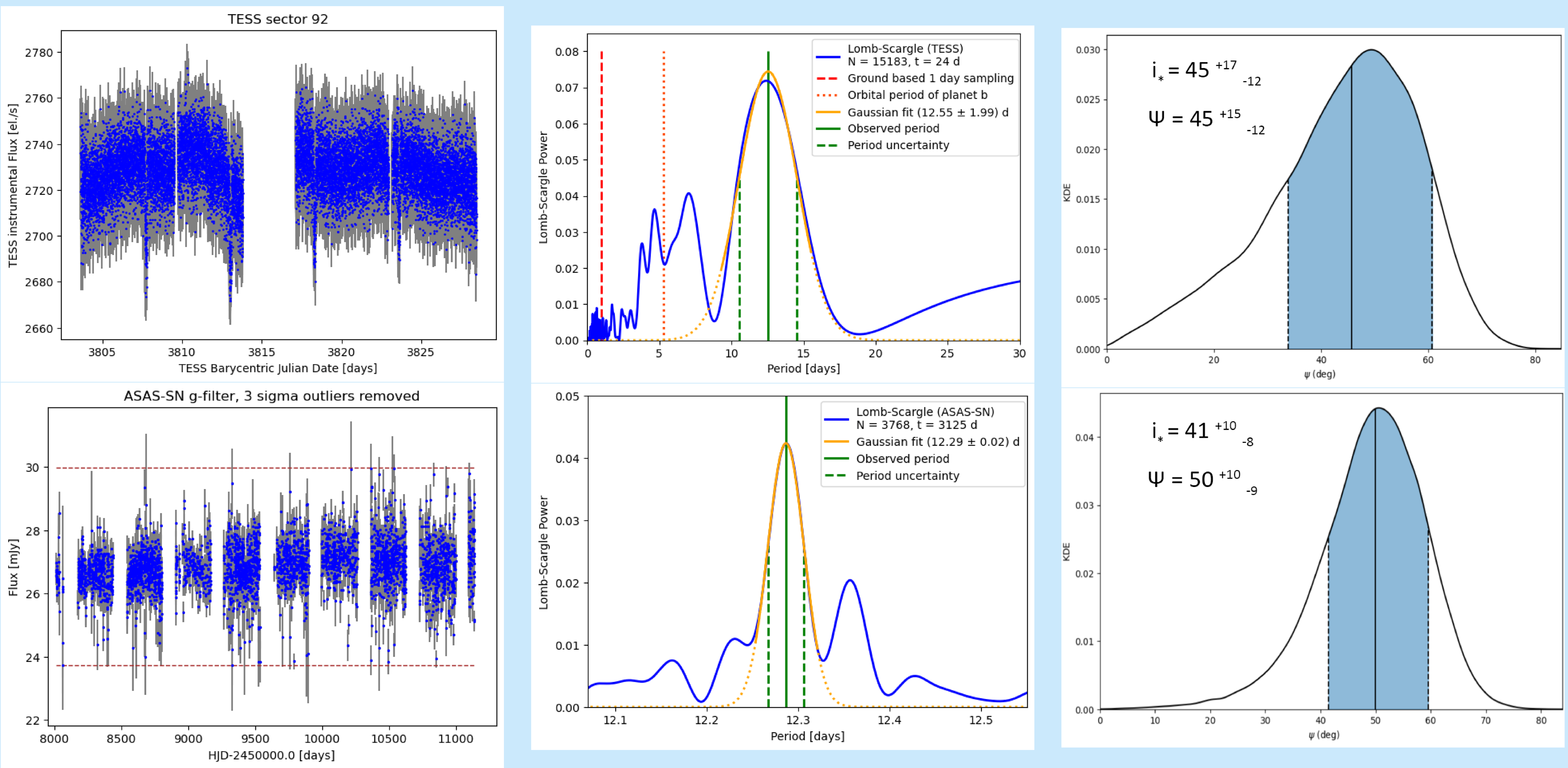
- ***TESS* photometry** [12] - better precision, but short photometric series, problems with contamination and systematics in data → initial estimates
- **Ground-based surveys** - day/night cycle and other aliases, significantly worse precision, but long photometric series available → precise periods
- Multiple survey projects available: ASAS-SN [13], KELT [14], Rubin Observatory [15], …
- Cross-check for systematics and seasonal changes of the periodogram
- Robust confirmation of rotational variability can be done by identifying period harmonics in the periodogram, but this is rarely possible

5) Goals

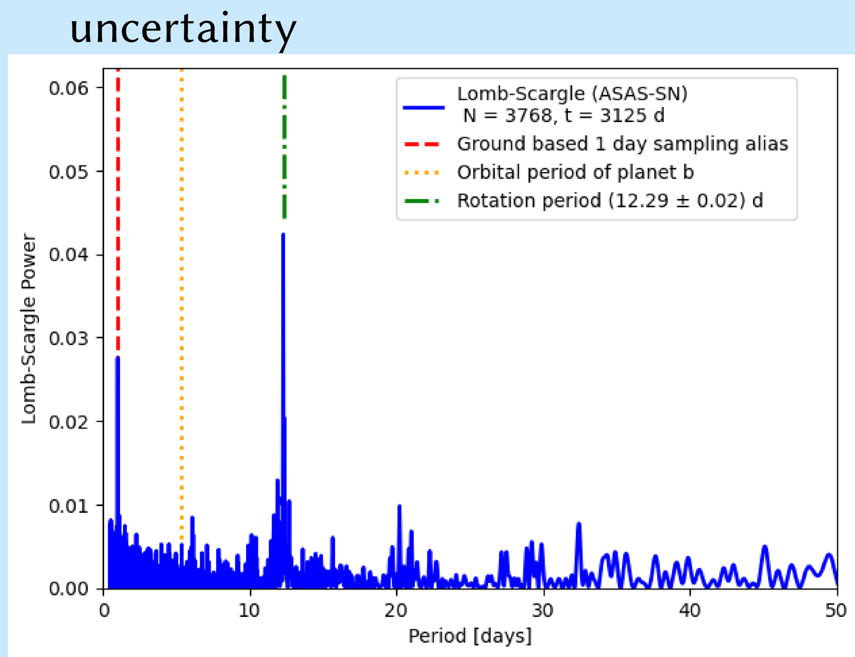
- Create the first homogeneous catalogue of rotational periods for systems with known spin-orbit angles
- Increase the sample of true spin-orbit angles by identifying new rotation periods and by measuring new sky-projected spin-orbit angles
- Compare the improved sample with theoretical models to characterize the dominant pathways in exoplanet evolution [1]

6) Proof of concept

The example is based on EPIC 219388192 system [16], with $v\sin(i_\star)$ artificially lowered to (2.6 ± 0.5) km/s for a better illustration of the effects of period uncertainty in a case of a possibly misaligned planet. Light curves from TESS and ASAS-SN were analysed using Lomb-Scargle periodograms [17, 18, 19]. From the resulting rotation periods, stellar inclination and true spin-orbit angles were determined using the *coPsi* Python package [9, 20].



- Precise values and uncertainties of the period were derived from Gaussian fits of the dominant peaks
- The ≈130 times longer time series from ASAS-SN leads to ≈100 times smaller period uncertainty, despite the significantly worse precision of individual brightness measurements
- This difference results in ≈30 % improvement in true spin-orbit uncertainty



7) Future

- Gaia DR4 will provide improved stellar parameters & orbit-orbit angles for binary systems
- Cross-check with rotational periods from TESS All-Sky Rotation Survey [21]
- Test alternative period analysis methods
- Include data from other photometric surveys
- Compare the obtained distribution of true spin-orbit angles with results from evolutionary models

8) References

1] Albrecht et al. (2022), Stellar Obliquities in Exoplanetary Systems
2] Rossi et al. (2026), True spin-orbit obliquity distribution: Data-driven confirmation of no clustering of misaligned planets
5] Biddle et al. (2025), One-third of Sun-like stars are born with misaligned planet-forming disks
6] Skarka et al. (2022), Periodic variable A-F spectral type stars in the northern TESS continuous viewing zone. I. Identification and classification
9] Masuda, K. & J. N. Winn (2020), On the Inference of a Star's Inclination Angle from its Rotation Velocity and Projected Rotation Velocity
16] Zak et al. (2025), Ten Aligned Orbits: Planet Migration in the Era of JWST and Ariel
19] VanderPlas, J. Understanding the Lomb-Scargle Periodogram

Full references available online

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